

A call to integrate animal movement into biodiversity indicators

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Abstract

Animal movement data have transformed our understanding of ecological systems and shaped conservation practice, but have limited influence on tracking progress towards international biodiversity goals. Existing biodiversity indicators adopted in frameworks such as the Kunming–Montréal Global Biodiversity Framework – the primary multilateral conservation agreement that aims to halt and reverse biodiversity loss by 2030 – are typically not responsive enough to detect biodiversity change in time to guide action, nor sufficiently biologically informative to explain or predict changes. In this Perspective, we provide seven reasons why movement data can help to tackle these limitations by adding biological realism, mechanistic understanding, and early-warning capacity to current and future indicators. Movement data already inform conservation efforts, from local management to global treaties for migratory species, and are increasingly helpful to uncover sources of environmental change, while enhancing monitoring capability and policy relevance. We recommend that the scientific community ground existing connectivity metrics in empirical movement data, develop new indicators that flag rapid change, and invest in modelling and attribution studies that use movement data to identify drivers of biodiversity loss and recovery.

Sections

Introduction

The role of biodiversity indicators in conservation policy

Current biodiversity indicators and their limitations

Why biodiversity indicators based on movement data are needed

Outlook

Introduction

The movements of individual animals have been monitored for scientific research since 1890, and the resolution of data and the number of species observed has steadily increased as methodologies have become more advanced^{1–3}. A suite of technologies, from place-based sensors (such as *Motus*) to tracking devices fitted on animals that communicate with satellites, are capable of automatically recording the location of individual animals repeatedly through time^{1,4}. The rich information on animals' fine-scale behaviours and interactions with their surroundings offered by movement data has unlocked new ecological insight into the causes and consequences of animals' movements⁵, markedly improving our ability to understand and predict wildlife responses to rapid environmental change⁶. Movement data have informed conservation policy across a range of scales^{4,7–10} and have contributed to numerous conservation applications, including identifying critical habitats and areas of ecological importance¹¹, designing marine protected areas¹², siting conservation easements⁹, informing species status assessments for conservation planning¹³, and contributing to multilateral agreements to reduce bycatch¹⁴.

Despite these advancements, movement data are not yet well integrated into tracking progress towards specific conservation goals and targets – the most prominent of which are the ambitious 2030 Targets and 2050 Goals to protect biodiversity set forth by the Kunming–Montréal Global Biodiversity Framework (GBF)^{15,16}. This framework, adopted under the UN Convention on Biological Diversity, aims to improve the state of biodiversity globally and both encourages and is mirrored by the rise in corporate and financial biodiversity assessments^{17,18}.

Biodiversity indicators are core to both meeting conservation aims and biodiversity assessments, as they translate data into quantitative insights about the pace of biodiversity change and whether we are moving towards or away from desired targets¹⁹. Indicators are valuable for shaping policy and interventions because they provide standardized and comparable values across contexts. Creating indicators that are generally applicable is a substantial challenge and movement data have largely not been integrated into indicator development, probably owing to logistical and methodological complexities in incorporating them²⁰. However, as movement data can provide important insights into biodiversity change, their absence limits the full potential of indicators to convey the best possible insights to guide conservation actions.

In this Perspective, we argue that movement data can and should be embedded within biodiversity indicators, particularly for those used within the GBF. We first describe key existing indicators and their limitations, and then present seven reasons why movement data can help to address some of these shortcomings (Fig. 1). Finally, we outline challenges to incorporating movement data and recommend a path forward that identifies urgent priorities and forward-looking developments for translating movement data into actionable biodiversity indicators. This work, and development of the seven reasons, arose from interdisciplinary discussions through the *Animal Movement Biodiversity Observation Network* (Move BON; established in 2025), operating under the *Group on Earth Observations Biodiversity Observation Network* (GEO BON) (Supplementary Information).

The role of biodiversity indicators in conservation policy

Indicators measure the change through time of a component of biodiversity relative to a baseline²¹. This 'component of biodiversity' is

typically an essential biodiversity variable (EBV)²², such as species abundance, species distribution or community composition, which are standard data products that can be summarized with indicators through time and space. Indicator values must be responsive to interventions and should ultimately show improvement as targets are reached¹⁹. Tracking biodiversity change requires indicators whose values respond after the subject of interest changes (termed 'lagging indicators'; for example, the Biodiversity Intactness Index²³ reflects past species abundance or richness losses), alongside indicators whose values change before the subject of interest (termed 'leading indicators'; for example, the Red List Index²⁴ tracks the risk of future extinctions)¹⁹.

Biodiversity indicators were conceived over three decades ago as a way to influence environmental policy²⁵. Since then, many biodiversity indicators have arisen from a growing need to monitor progress towards conservation targets, starting with the 2002 World Summit of Sustainable Development's objective to slow biodiversity loss by 2010²¹. This objective sparked a period of indicator development during the 2000s, leading to species distribution and abundance indicators in Europe^{26,27}, and global indicators such as the Red List Index²⁴, the Living Planet Index²⁸ and the Biodiversity Intactness Index²³. However, these global conservation targets did not come with a mandate to report progress with indicators, which is widely considered to have contributed to the failure of the 2020 Aichi targets²⁹. To achieve future targets, indicator-based monitoring is now the focus of the GBF.

Parties to the GBF are required to report a set of mandatory 'headline' indicators to track progress in their National Biodiversity Strategies and Action Plans, which can be supplemented by optional 'component' indicators to cover elements not captured by the headline set, and 'complementary' indicators for thematic or in-depth analysis of goals and targets^{29,30}. All headline indicators must adhere to a set of requirements, including requirements for the availability of relevant data and their ability to be measured relative to a baseline and calculated for entire nations, among other conditions. However, the suite of indicators in the GBF are neither exhaustive nor perfect. These indicators are part of a broader set of metrics that, although they might not fit the GBF criteria exactly, are essential for tracking and understanding the complex problem of biodiversity change, and could support future indicators beyond the GBF³¹.

Current biodiversity indicators and their limitations

The GBF monitoring framework includes a suite of indicators³⁰ (CBD Decision 15/5, table 1) to monitor progress towards biodiversity targets. Among the many headline, component and complementary indicators, five widely applicable indicators that are directly relevant to animal species and ecosystems can be used to illustrate broader limitations of current indicators. Collectively, these five indicators represent decades of development and are critical for measuring biodiversity change, but each carries limitations that movement-based indicators can help to address (Fig. 2).

First, the proportion of populations within species with an effective population size greater than 500 ($N_e > 500$) is an indicator for monitoring genetic diversity loss that can anticipate population declines³². However, this indicator cannot directly capture the functional connectivity – the actual movement of individuals between populations – that sustains genetic diversity. Second, the Living Planet Index tracks trends in the relative size of vertebrate populations and can flag widespread vertebrate population declines²⁸, but only after these declines are already underway and often without the spatial

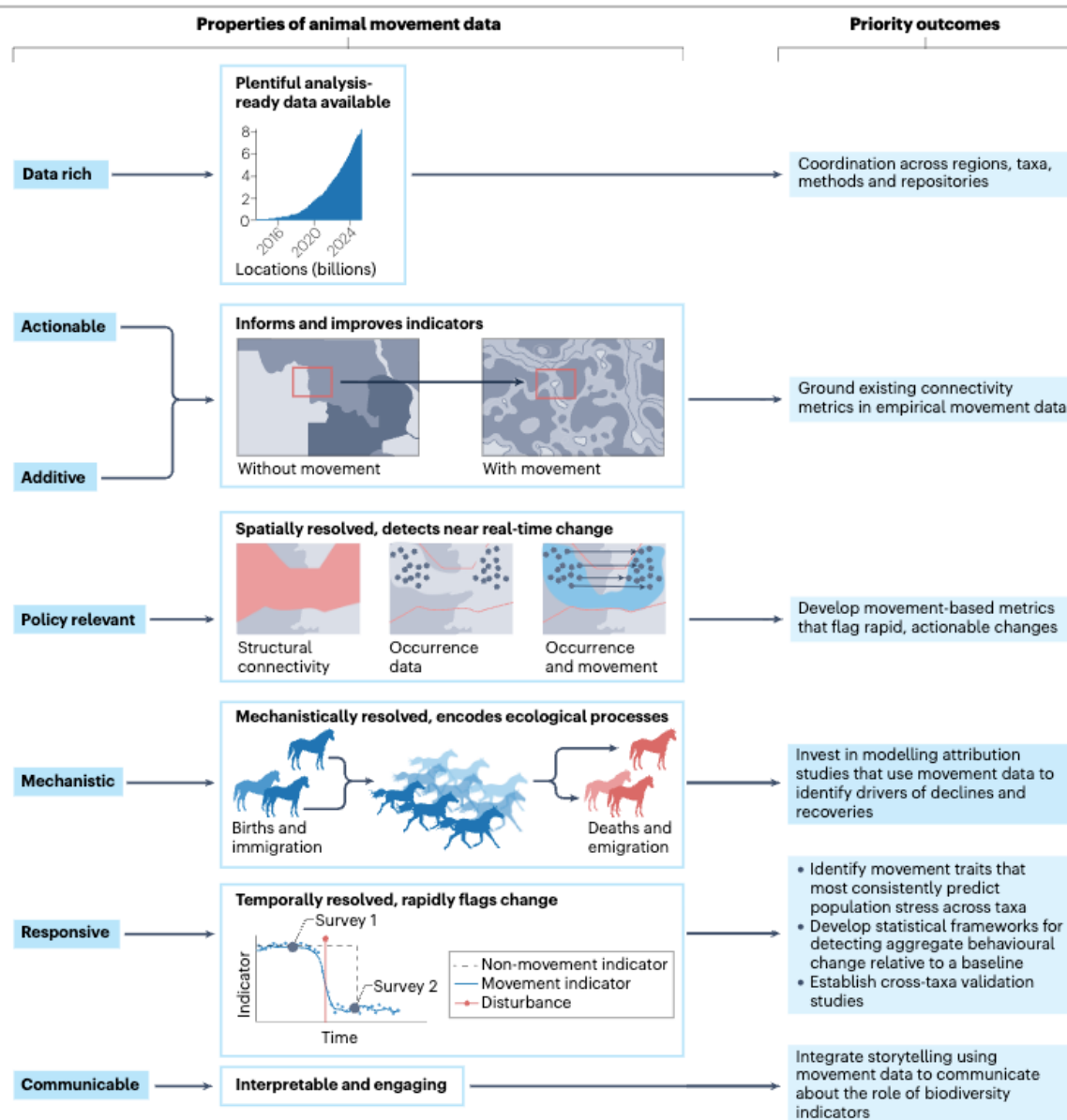


Fig. 1 | Movement data should be incorporated into biodiversity indicators. Owing to improvements in technology and data sharing, movement data are more available than ever before to inform policy and engage the public in

protecting biodiversity. By providing a rich time series of repeated observations of individual animals, movement data capture ecological processes that can inform a wide range of indicators.

resolution or demographic mechanism needed for attribution or local conservation action³³. Third, the Biodiversity Intactness Index, which tracks ecosystem integrity as an estimated proportion of original species abundance remaining in human-modified landscapes, measures ecosystem intactness relative to an intact baseline^{23,34}. This indicator can identify declines and local extinctions, but only after these losses have occurred³⁵. Fourth, the Species Habitat Index tracks changes in the extent and condition of species habitats³⁶, and relies on species occurrences and coarse land cover to define habitat; this data only poorly captures fine-scale habitat selection, particularly for highly mobile species^{37–39}. Finally, the Red List Index, which tracks changes in species' extinction risk status²⁴, can detect these changes in extinction

risk status after substantial, sustained declines have occurred; by the time a species has been reassessed to a higher risk Red List category, opportunities for cost-effective intervention might be limited.

These gaps reflect fundamental mismatches between current indicators and conservation aims. Current indicators largely measure late-stage outcomes of biodiversity loss⁴⁰ as they are unable to detect ongoing change, and can provide only limited insights into mechanisms or drivers of population change as they incorporate insufficient data at the relevant spatial scales or subpopulation level. By contrast, proactive conservation requires early detection of the processes driving loss at the spatial scales where management can act. Movement data offer a direct path to addressing important limitations

Perspective

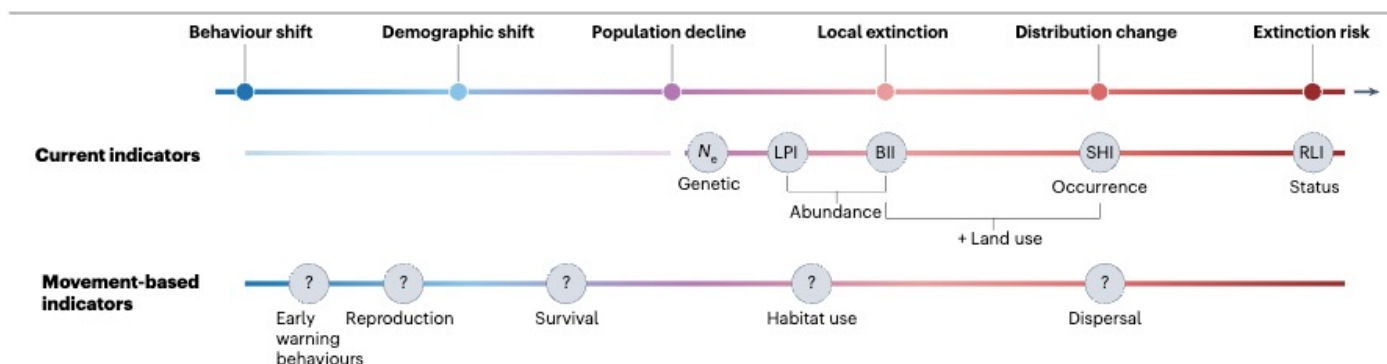


Fig. 2 | Movement data can help to create indicators that fill key gaps in biodiversity monitoring. Existing biodiversity indicators are informed by data on species abundance ($N_e > 500$, Living Planet Index (LPI)), land use and species abundances or occurrences (Biodiversity Intactness Index (BII) and Species Habitat Index (SHI)) or extinction risk status of species (Red List Index (RLI)) – data that are signals of a collapse that is already underway. Effectively halting biodiversity loss requires indicators that target earlier phases of biodiversity

collapse when interventions are still possible and most effective. Movement data can motivate the development of new indicators (illustrated by question marks) that measure early shifts in behaviour and demography (such as reproduction and survival), as well as later signals of collapse (such as reduced habitat use and dispersal) to support decisions and conservation actions along the full timeline of biodiversity collapse.

of existing indicators and critical gaps in biodiversity monitoring: they are spatially explicit, can be collected autonomously and can reveal behavioural shifts⁴⁰ before population-level signals emerge. So far, none of the GBF headline indicators are based directly on movement data, despite the potential of movement data for measuring EBVs (Box 1) and their long legacy of impact on conservation through other policy mechanisms^{4,7–10,13}.

Why biodiversity indicators based on movement data are needed

Movement data can help to address limitations of current indicators, both within the GBF monitoring framework and more broadly. This section presents seven key arguments for integrating movement data into biodiversity indicators, including the unique richness of movement data, the long-term potential to create new indicators that are more responsive and actionable than current indicators, the near-term capability to improve existing indicators, the demonstrated relevance of movement data to conservation policy, the ability of movement data to provide understanding of the mechanisms driving changes, the future potential to develop early-warning systems using movement data, and the ability for movement data to engage and inspire the public (Fig. 1).

Movement data are a uniquely rich source of information for indicator development

One of the most crucial requirements of an indicator is the availability of appropriate supporting data. Movement data have long been skewed towards large-bodied animals in well-resourced areas^{41–43}, meaning consistent metrics or comparisons across regions or through time were previously not possible. As GBF headline indicators must be able to be calculated for all nations, their ability to draw on movement data has so far been limited²⁰. However, movement data are one of the most rapidly expanding forms of biodiversity data⁴⁴. Decades of automated tracking have now produced vast datasets and data collection has expanded exponentially in the past 20 years⁴⁵, capturing movement across terrestrial, aerial and aquatic domains^{46,47}. Although movement data continue to suffer from taxonomic, geographic and ownership biases⁴² (see ‘Challenges and limitations’), their potential for

widespread use is increasing as the cost and labour required for data collection fall, and as tracking devices can now be deployed on smaller organisms^{1,42,48} (for example, satellite-linked transmitters weighing 1 g are under development for the [International Cooperation for Animal Research Using Space](#) initiative). However, even as technologies improve, consideration of impacts on animal welfare must be central to any new data collection effort^{49,50}.

Movement data are unique in providing a tremendous volume of repeated observations on individual animals. For example, [Movebank](#), one of the most popular repositories for storing movement data, now contains over 10 billion records of the locations of over 300,000 individual animals. By comparison, the Global Biodiversity Information Facility currently stores approximately 3.1 billion records across the entire tree of life for an unknown number of individual organisms. The density of information for each individual animal contained within movement data (such as for tracked individuals) is promising for developing indicators that detect changes more rapidly than current indicators – at least for a small but increasing subset of species. Recognizing the power of these data for many applications, major national and international networks have worked to coordinate data collection, standardization and aggregation (for example, the [Ocean Tracking Network](#)⁵¹, the [BirdLife Seabird Tracking Database](#)⁵², the [Shorebird Science and Conservation Collective](#)⁹ and the [International Bio-Logging Society](#)^{47,53}).

Movement data can support new, actionable, model-based indicators

Biodiversity indicators for the GBF must be responsive enough to be evaluated and track progress within the 2030 and 2050 timelines (for Targets and Goals, respectively)^{19,54,55}. Ideally, indicators should also be ‘actionable’ – that is, they enable interventions that have measurable effects^{19,56}. However, most existing indicators lack the spatial resolution and responsiveness needed to inform timely local action because they are designed to report progress towards national and international targets over 5–10-year timelines (or longer)⁵⁵. For example, Red List assessments incorporate early signals of population declines to determine extinction risk²⁴, and reassessments are conducted at least every

10 years⁵⁷. Beyond data constraints, existing species-focused indicators are designed to measure broad-scale biodiversity patterns from population declines up to global extinction risk (Fig. 2), missing finer-scale changes in behaviour, physiology, morphology and demography that often precede the collapse of wildlife populations⁴⁰.

Movement data could, in principle, address shortcomings in the responsiveness and actionability of indicators. As most movement data are 'born digital'⁴⁴ – that is, collected automatically and increasingly live-streamed to shared data portals (such as Movebank) – they could support indicators that are updated frequently, greatly reducing reporting lags⁵⁸. Movement data also comprise large volumes of spatially explicit records with precise timestamps, enabling insight into how wildlife engage with their surroundings. This combination of temporal and spatial resolution opens the door to indicators that are more actionable than current indicators, responsive enough to detect changes that precede population decline, and precise enough

to attribute observed trends to specific habitats, pressures or interventions. In addition to movement data, similar potential exists for other fast-paced data types (for example, accelerometer, magnetometer and bioacoustic data, which increasingly are collected together with movement data⁵⁹; camera traps; and citizen science data) and indicator construction (for example, aggregation and averaging) to also capture early signals of biodiversity change.

In the short term (within the next 5–10 years), model-based indicators (Fig. 3) derived from EBVs based on movement data could be developed (Box 1). Biodiversity responses of interest – such as movement traits⁶⁰, home range use, health proxies and mortality rates – can be estimated from relatively sparse tracking datasets and then related to dynamic environmental drivers⁶ that are measured consistently across space and time via remote sensing (such as land cover, temperature and vegetation structure). This enables movement data to be leveraged well beyond the locations and/or time frames that animals were actually

Box 1 | Potential for quantifying essential biodiversity variables with movement data

Movement data can support a broad range of essential biodiversity variables, some of which could lead to new indicators or improvement in indicators with future developments.

Genetic composition indicators

Combining genetic and telemetry data can support functional connectivity indicators (for example, by identifying barriers to gene flow) or population health (by identifying movement between at-risk populations). Existing genetic diversity indicators³² could eventually be combined with movement data to track early warnings of population declines, such as behavioural signals of genetic diversity losses (Box 2).

Species populations

Movement data can be used to define populations^{136,137}, which can support existing indicators of extinction risk – informing indicators such as the Red List Index²⁴ and connectivity-based indicators (Box 2).

Species distributions

Movement data can refine species–environment associations to more precisely define critical habitats^{138,139} and to up-scale individual movement for improved predictions of species distributions^{6,75}. These improvements could help to support indicators that rely on species distribution estimates (such as the Species Protection Index¹⁴⁰) or habitat associations defined by intersecting expert range polygons with land cover associations (such as the Species Habitat Index³⁶) – or to develop a new indicator to track species' home range use (Fig. 3).

Species abundances

Increasing potential exists for better estimates of abundance from movement data; for example, using radar data for assessing insect biodiversity¹⁴¹ and satellite data to track seabird abundance in foraging grounds at sea¹⁴² – but major challenges remain. When these challenges are resolved, movement data could contribute to existing indicators of abundance (such as the Living Planet Index³³)

or inform new indicators with attribution power (for example, mechanisms of mortality¹⁰⁵).

Species traits

Movement informs many species-level traits, such as thermal physiology¹⁴³, reproduction success⁹⁸, phenology (migratory timing and compensatory behaviours)^{144,145}, displacement distance, range size, diurnality and behavioural modification¹⁴⁶, even in remote locations for at-risk species¹⁴⁷. The relationship between movement and species traits offers much potential for indicator development, particularly if movement data are combined with mechanistic models¹⁴⁶ or correlative models¹⁴⁸ linking traits with attribution to estimate indicators that provide mechanistic insights about biodiversity change (Box 2).

Community composition

Movement data can be used to estimate diversity¹⁴¹ and functional diversity, and are fundamental for many types of interactions (seed dispersal, disease spread and predator–prey). Many possibilities exist for new indicators or for informing more reactive community-level indicators than currently exist¹⁴⁹ to track composition changes at a faster pace.

Ecosystem structure and functioning

Movement data can be unpacked to track phenology (such as shark movement predicting upwelling¹⁵⁰, crop phenology predicting bird movement¹⁵¹, geese responses to wildfire¹⁵² and vegetation use of deer¹⁵³) and ecosystem structure (thanks to height and/or depth information)¹⁵⁴. These reactive data could feed into existing indicators, some of which are not as closely related to biological responses. For example, a metric of ecosystem intactness (Biodiversity Intactness Index²³) or the single adopted indicator for B.1 'Services provided by ecosystems' could be informed by movement responses to ecosystem fragmentations. New indicators could also be derived to monitor ecosystem structure and functioning, such as the presence of taxa that indicate impending changes, restoration success or disturbance from known stressors (Fig. 3).

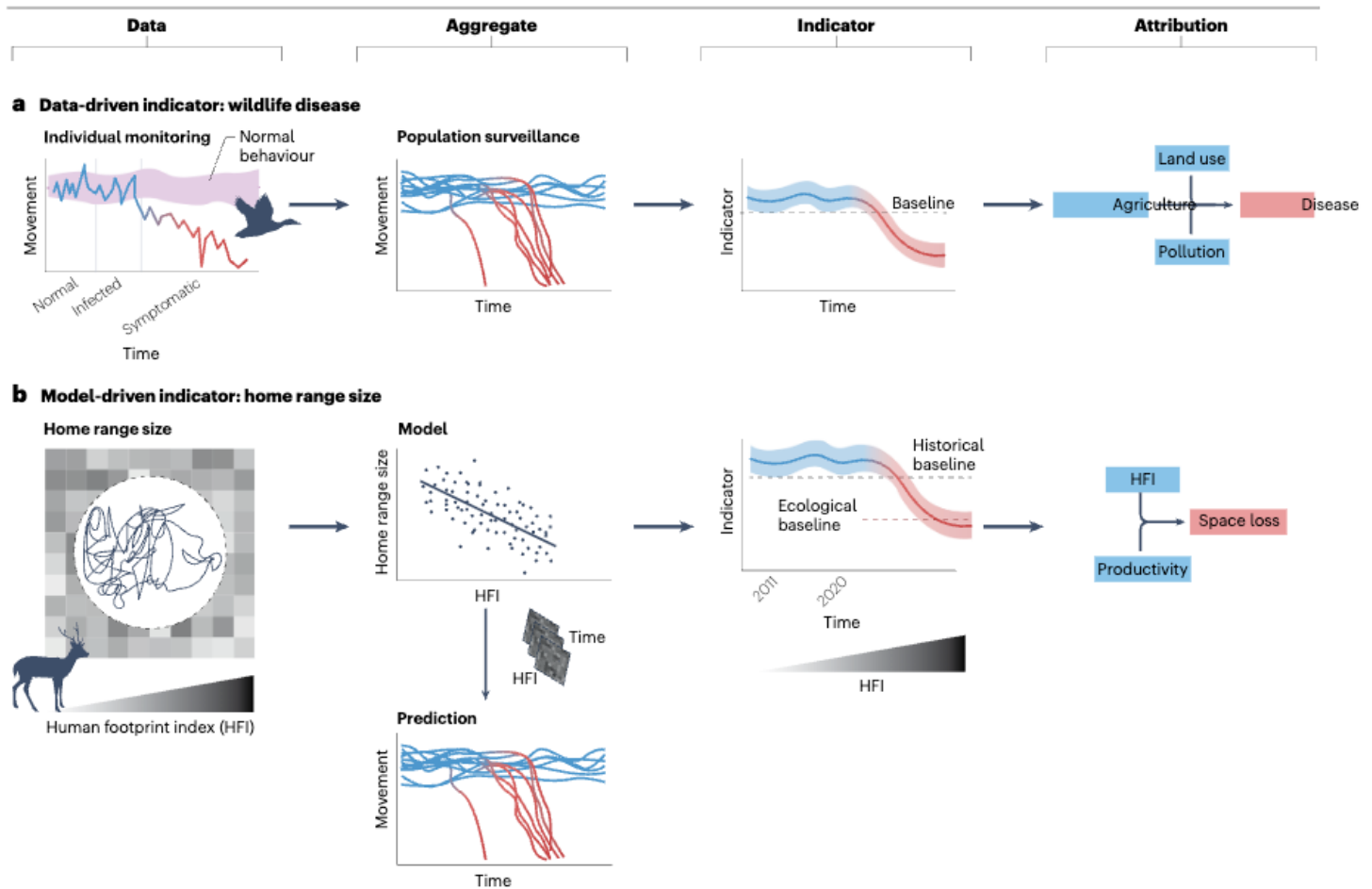


Fig. 3 | Creating biodiversity indicators with movement data. Movement data could be used to create biodiversity indicators through two different analytical approaches that could be applied to a wide range of indicators – a data-driven approach or a model-driven approach. **a**, A data-driven approach requires collecting movement data continuously to estimate the indicator over time. For example, movement data can be used to detect changes in the health status of individual animals. An indicator could therefore aggregate and summarize health status over time, based on movement data that are continuously being collected for a population (which is already possible for some closely monitored populations¹¹³). This indicator could be used to attribute zoonotic disease transmission to a range of potential drivers to inform interventions. **b**, By contrast, a model-driven approach leverages estimated relationships between

response variables and an environmental driver to predict change through time. For example, a model could be used to estimate the relationship between home range size (estimated from movement data^{130–132}) and an aggregated measure of the human footprint to create an indicator of animals' use of their home ranges^{133–135}. Instead of continuously collecting movement data, the model could be used to predict and potentially forecast the indicator over time based on changes in human footprint. The indicator could then be compared with a historical baseline (that is, its value during the reference year) or to an ecological baseline (such as a threshold value informed by an ecological understanding of how much home range is theoretically needed for the species to persist). Adapted with permission from ref. 133, Wiley.

observed⁶¹, producing spatially continuous estimates of biodiversity change. This model-based framework also allows for tailoring statistical models specifically to management interventions and naturally deals with attribution (that is, biodiversity change is predicted based on driver variables). Model-based approaches offer a way to balance responsiveness and noise, as they are sensitive enough to detect trends while remaining robust to biologically insignificant short-term population fluctuations; however, work on benchmarks and validation will be necessary, especially when integrating newer artificial intelligence models with traditional statistics.

In the long term (beyond 10 years), movement-based indicators would be deployed at the scale and coverage required by the GBF;

however, accomplishing this within the current target window remains a substantial challenge. Tagging programmes are unevenly distributed across taxa and regions, and even rapidly growing datasets have large spatial and taxonomic gaps (see 'Challenges and limitations'). A movement indicator built solely on empirical tracking data, for example, is unlikely to be developed before 2030. However, future indicators could inform early-warning systems (see 'Movement data can enable early-warning indicators of rapid change' and 'Outlook').

Movement data can improve existing indicators

Movement data have the potential to support many of the EBVs that are tracked with existing indicators in the GBF monitoring

framework (Box 2). Their ability to refine the estimates of connectivity and species–habitat associations that underpin several indicators is particularly promising.

Protecting habitat connectivity is central to the GBF because it is essential to the long-term persistence of biodiversity and forms a key part of Target 3, which aims to achieve a ‘well-connected’ protected area network covering 30% of land, waters and seas. Currently, several non-headline indicators, such as the Protected Connected Index and Protected Area Representativeness and Connectedness Indices, estimate structural connectivity using landscape characteristics, but do not capture functional connectivity or species persistence^{62–64}. To more directly assess the functional connectivity of protected areas, the Protected Area Isolation index, currently a complementary indicator, leverages movement data to relate the movement of animals to human disturbance⁶⁵.

Further uses of movement data could improve current indicators (Box 2) by directly integrating predictions of connectivity from observed movements to validate, understand, and protect the true

rates of interchange between protected areas^{66,67}. In addition, movement data allow dynamic visualizations to directly map the current distribution of tracked individuals and their flow through landscapes and seascapes, such as during dispersal events and migration^{68–72}, to better engage policymakers and increase public support for conservation and policy actions (see ‘Movement-based indicators can inspire public engagement’).

Movement data are also poised to improve indicators derived from species distributions (for example, Species Habitat Index and Biodiversity Intactness Index). Currently, species distributions are primarily estimated by linking observed species’ occurrences with climate and habitat data to model the suitability of locations across space and time⁷³. However, species’ distributions are comprised of the movements of individual organisms, which can exhibit complex relationships with underlying environments and constraints on movement that are not well described by correlative niche models^{6,37–39,74,75}. In particular, species distribution models for highly mobile and migratory species could be improved by explicitly including movement

Box 2 | Examples of where movement data could enhance existing indicators to better track Global Biodiversity Framework targets

Movement data can contribute insights relevant to many existing indicators³⁰, ensuring that they provide more effective and frequent estimates of progress towards the various outcomes sought by the the Kunming–Montreal Global Biodiversity Framework Targets than are currently available.

Target 2: ecosystem restoration

Example current indicator

Headline indicator 2.2 (Area under restoration). This metric quantifies effort in terms of the extent of land area in preparation, in the process or in completion of being restored, but not ecological success.

Contribution of movement data

Movement data provide a powerful tool to monitor restoration outcomes, such as the rate of recolonization of restored sites by mobile species; whether restored patches are integrated into the wider landscape network by serving as stepping stones; and the return of animal-mediated functions such as pollination or seed dispersal. This would elevate indicator 2.2 from a simple measure of restored area to a more meaningful metric of functional restoration.

Target 3: protect and conserve areas

Example current indicator

Headline indicator 3.1 (Coverage of protected areas and other effective area-based conservation measures). This indicator is a measure of area. Even advanced structural metrics such as the Protected Area Representativeness and Connectedness Indices⁶⁴ and Protected Connected Index⁶² rely on modelled probabilities of movement.

Contribution of movement data

Trajectory analysis of animals provides a direct empirical test of the ‘well-connected’ clause in Target 3. By overlaying movement

paths on landscapes, habitat corridors that are actually used can be documented, cryptic barriers (for example, roads, fences, human activity hot spots) that render structural connections impermeable can be identified, and the true rate of interchange between protected areas quantified. These insights enhance indicator 3.1 by measuring functional connectivity, allowing managers to assess whether their protected area network is functioning as intended, not just if it meets an area target.

Target 4: species recovery

Example current indicator

Headline indicator A.4 (Proportion of populations with effective population size (N_e) >500). This indicator tracks extinction risk and genetic health, often based on slow-moving population and genetic trends.

Contribution of movement data

Although genetic sampling provides the ultimate measure of genetic health, movement is the physical mechanism that facilitates gene flow. Movement data provide a critical, often more accessible, proxy for estimating and managing for indicator A.4. Using movement data, the rate of immigration and emigration — key parameters in genetic models that estimate N_e — can be quantified to determine whether fragmented populations function as a larger, interconnected metapopulation. Even a few successful immigrants per generation can greatly increase a small population’s N_e and prevent inbreeding depression (by ‘genetic rescue’)¹⁵⁵. Therefore, movement data enable a series of small populations (each with N_e <500) to be classified as a single, functional genetic unit (with a collective N_e >500) if sufficient interchange is empirically documented. This transforms the indicator from a static attribute into an actionable management target: instead of waiting for genetic degradation to occur, we can use movement data to proactively identify and protect the specific corridors that sustain genetic diversity.

data⁷⁵ to provide a more refined understanding of the underpinning animal–environment relationships (Box 1).

A particularly promising avenue for improving existing indicators is the integration of behaviour-specific habitat selection models with movement data. Advanced analytical methods allow movement data to be partitioned by behavioural state – for instance, foraging, transiting or resting – and combined with resource selection functions or step selection functions to describe how animals use their environment during specific activities^{76,77}. This approach goes beyond static habitat suitability maps to reveal, for example, which habitat types are critical for foraging versus migration, and how this selectivity changes across seasons or in response to disturbance. For indicator development, behaviour-specific resource selection functions and step selection functions offer a pathway to increasingly biologically realistic estimates of critical habitat – grounding the Species Habitat Index and related indicators in empirically observed animal decisions rather than modelled associations with climate or habitat. More broadly, movement data can be scaled up^{6,61} and could inform and improve a suite of EBVs and essential ocean variables related to species' abundances, phenology, and physiology as well as community composition, ecosystem function, and ecosystem structure^{22,78,79} (Box 1).

Movement data already inform conservation policy

Indicators are one of the few conservation policy mechanisms into which movement data have yet to be integrated. The impact that movement data have had within conservation policy is strong evidence that they are well suited and ready to inform indicators within the GBF¹⁰. A particularly salient example is the substantial role of movement data in informing international coordination of species protections.

The movement of animals irrespective of political boundaries is explicitly reflected in the Convention on the Conservation of Migratory Species of Wild Animals (CMS), a United Nations treaty that supports international coordination to conserve migratory species. Movement data provide unique information on land and ocean transboundary dispersion and migration of animals^{68,71,72,80–83}, and have therefore had a critical role in informing international treaties⁸⁴. Following the mandate from the Conference of the Parties (COP 11, 2014), CMS developed an [Atlas on Animal Migration](#) and updates (COP 12-14) based on movement data, and hosts key data synthesis initiatives including the [Eurasian African Bird Migration Atlas](#), the migration maps from the [Global Initiative on Ungulate Migration](#)⁸⁵, and the [Atlas for the Americas Flyways](#). CMS also formally recognizes Global Flyways and has an established programme of work⁸⁶. For example, the African-Eurasian Migratory Bird Agreement (AEWA, a daughter agreement of the CMS) coordinates international conservation action for migratory waterbirds across the African-Eurasian Flyway⁸⁷, including identifying a Critical Site Network and establishing task forces to address specific threats (such as illegal killing, taking and trade of migratory birds). Movement data have also been used to delineate important sites at a population level that can be compared against the internationally agreed criteria for Key Biodiversity Areas⁸⁸. Key Biodiversity Areas are globally important sites for the persistence of biodiversity and have had widespread impact in informing protected area designation⁸⁹, including the North Atlantic Current and Evlanov Seabasin Marine Protected Area that was originally identified from an analysis of seabird tracking data⁹⁰.

Outside formal conventions, numerous initiatives have leveraged the power of movement data to motivate conservation across borders. Examples include the Atlas of North America's Migratory Birds⁹¹, the [Migration Connectivity in the Ocean System](#), [Ocean Biodiversity](#)

[Information System Spatial Ecological Analysis of Megavertebrate Populations](#), [Global Flyways](#), the [Shorebird Science and Conservation Collective](#)⁹ and [Global Wader](#). These initiatives provide timely and actionable information to management and governance entities to meet ecological connectivity targets across political boundaries^{68,71,81,85}. Further, this body of work clearly indicates that movement data are effective and valuable for informing policy frameworks similar to the GBF¹⁰.

Movement data capture the underlying mechanisms driving biodiversity change

Indicators can only effectively support conservation action if observed changes are attributable to underlying biological mechanisms that can be used to design mitigation efforts⁹² – a substantial challenge for most indicators. Movement data, and other individual-based data types, could support indicators that embed the underlying causal mechanisms driving observed changes (Figs. 2 and 3). For example, movement data can be used to directly measure component processes of population change – such as survival^{93–96}, reproduction^{97,98}, immigration and emigration^{99,100} – and their context, which can generate deeper insights into 'black box' declines (for which the mechanisms driving change are unknown) in population abundance recorded by current indicators¹⁰¹.

Movement data enable linking behavioural responses to environmental change to population-level and species-level processes⁶. Doing so can reveal nuanced and sometimes counterintuitive patterns that would be otherwise invisible. For example, climate-driven variation in precipitation on Caribbean non-breeding grounds has been shown to shape the migration phenology and, in turn, reproductive success and dispersal patterns of American redstarts (*Setophaga ruticilla*)^{99,100,102}. These individual, environmentally mediated movement decisions ultimately scale up to population-level variation in annual survival rates that has led to a ~500-km southward shift in the northern breeding range of this species over the past 30 years³⁷. This outcome runs counter to the general prediction that climate change will push species polewards¹⁰³ and could only be revealed with the detailed and repeated observations of individuals offered by movement data.

Similarly nuanced effects of individual movement behaviours probably occur across a diversity of taxa on land and in marine environments¹⁰⁴, but remain largely underexplored. Failure to incorporate the mechanistic insights afforded by movement data risks misunderstanding key spatial and demographic processes that underpin the dramatic changes in species' abundance, ranges and extinction risk observed globally¹⁰¹.

Movement data can enable early-warning indicators of rapid change

Creating indicators that detect ecological stress before populations begin to decline would fill one of the most important gaps of the GBF and most other multinational monitoring efforts, but is probably a longer-term ambition beyond the immediate 2030 deadline for the GBF targets. However, a suite of emerging data and metrics that could serve as precursors or supplements to formal indicators should be explored^{19,56}.

By capturing changes in animals' behaviours, near real-time movement data provide early-warning signals of rapid environmental, anthropogenic and epidemiological changes^{105–108}, long before traditional surveys¹⁰⁹. Shifting behaviour is often the first way that wildlife respond to environmental stress and physiological change¹¹⁰. Whether

the driver is natural, anthropogenic or disease-related, behavioural responses can reveal early signs of ecological disruption – often before population-level consequences are detectable by current indicators (Fig. 2). However, identifying whether behavioural changes are evidence of population declines to come or successful adaptation to external stressors remains an ongoing challenge. Either way, new indicators that capture behavioural responses by integrating movement data could provide an early-warning system for ecological changes as they emerge⁵⁸ (Fig. 3). For example, infectious diseases are an important driver of animal behaviour and can be detected using movement data owing to its fine spatial and temporal scale^{111–113}. This information can be used to signal potential outbreaks as they emerge¹⁰⁵ (Fig. 3). Further, animals can rapidly change their movements in response to extreme weather events^{114,115}. For example, striped bass (*Morone saxatilis*) have exhibited novel migration behaviours and evacuated an estuary in response to extreme weather¹¹⁶.

Although implementing such early-warning systems as future biodiversity indicators might seem ambitious today, work can begin by establishing baseline data to inform monitoring efforts so that they become practical in the future. A first step in this direction is the **MoveTraits** database, which provides standardized and comparable measures of animal movement behaviours or traits (for example, daily displacements and range sizes) derived from movement data⁶⁰. As a ‘living’ database, it is designed to continuously grow alongside movement data shared in data platforms by calculating traits directly from individual-level data using consistent methodology. MoveTraits, together with real-time analytics platforms such as MoveApps¹¹⁷ and EarthRanger¹¹⁸, provide the data infrastructure on which future early-warning indicators could be built. The immediate priorities are identifying which movement traits most consistently predict population stress across taxa, and which sensors best record these traits; developing statistical frameworks for detecting aggregate behavioural change relative to a baseline, building upon the rich statistical literature on change-point detection^{119,120}; and establishing the cross-taxa validation studies needed before a behavioural metric could function as a reliable policy indicator.

Movement-based indicators can inspire public engagement

Indicators within the GBF are most successful when they are easy to interpret and spark interest outside of the scientific community. Animal movements easily capture the public imagination. The journeys and stories of animals are inherently compelling and can provide a powerful entry point for connecting people to the natural world. Countless initiatives, including the **Tour de Turtles**, the **Wyoming Migration Initiative**, Audubon’s **Bird Migration Explorer**, California’s **P-22** mountain lion campaign, **Fahlo**, **Panthera** and the **World Wildlife Fund** – which has over 2.6 million and 3.8 million followers on Facebook and Instagram, respectively – have used movement narratives to inspire a groundswell of public support for transformative conservation action¹²¹.

This public resonance is not incidental to the GBF, as Target 21 explicitly calls on Parties to ensure that the public are aware of and engaged with biodiversity, recognizing that their support is essential for generating the political will and resources needed to implement the GBF. Indicators created from movement data – following the journeys of individual animals that people already care about – offer a direct way to link the public to measurable progress on GBF targets, which is particularly powerful as optimistic stories have been shown to drive engagement with conservation actions¹²². Co-producing movement-based indicators with practitioners and stakeholders from

the earliest stages of development, rather than retrofitting, is critical for building the trust and relevance that successful indicators need^{123,124}.

Outlook

The science is clear – movement data capture fundamental aspects of biodiversity and ecosystem function^{106,125}, relate to a suite of EBVs (Box 1) and routinely inform on-the-ground conservation^{4,7,8,13}. Translating these diverse data streams into reliable metrics and policy-ready indicators is not straightforward and will require researchers and practitioners to triage opportunities by tractability and impact. Some of the approaches to leverage movement data for biodiversity indicators proposed here might not meet the 2030 deadline for the GBF Targets, but they are nonetheless worth pursuing as work continues towards mitigating biodiversity decline well into the future.

Challenges and limitations

Not all applications of movement data to indicators are equally ready to be implemented, and near-term opportunities and longer-term ambitions should be assessed based on tiers of readiness and future potential. We propose categorizing future activities into three tiers and provide example activities based on feasibility. The first tier comprises activities currently feasible within GBF reporting timelines: for example, extending the methods of connectivity indicators such as the Protected Area Isolation index⁶⁵ to include empirical movement data, developing migratory phenology tracking indicators for well-covered species, and achieving mortality detection for intensively tracked populations. The second tier of activities are feasible within 5–10 years with targeted investment: for example, cross-taxa movement trait indicators using MoveTraits infrastructure, model-driven indicators combining sparse movement data with satellite-derived covariates, and functional connectivity indicators for well-studied migratory systems. The third tier contains longer-term ambitions (beyond 10 years) that require methodological breakthroughs: for example, real-time behavioural early-warning systems applicable across taxa, population-scale mortality monitoring, and ecosystem function indicators derived from movement data.

Several challenges affect all three tiers. First, taxonomic and geographic biases mean that easily accessible movement data^{47,118,126} remain concentrated on large-bodied vertebrates and in North America, Europe and a small number of intensively studied systems elsewhere^{42,43}. Equity and representativeness are explicit GBF indicator requirements³⁰, and near-term development must be transparent about the taxa and regions it represents. Wildlife movements are, in many ways, inherently suited to addressing these biases: they connect high-resource and low-resource regions through migration⁶⁸, are widely monitored in biodiverse regions to support ecosystem management¹¹⁸, and can be integrated and analysed using existing infrastructure available for use across regions and species at no cost^{47,117}. Addressing underlying gaps will require investment in low-cost tracking technology, international partnerships and the data pipelines, which MoveBON is designed to provide. However, any study that involves the capture, handling and tagging of an animal must carefully consider the potential impacts on the animal’s welfare⁵⁰. Second, for all rapidly responding and most data-driven indicators, uncertainty exists regarding whether the same behavioural change can reflect adaptive response or impending decline, and distinguishing between these outcomes requires baseline data on fitness consequences and statistical frameworks that currently exist for very few species. As false positives risk unnecessary interventions, behavioural early-warning indicators should be treated as research priorities and preliminary

metrics rather than policy-ready indicators until these foundations are in place. Finally, current tracking technology is typically short lived relative to the lifespan of the species being tracked. To fulfil the vision of movement data as a comprehensive biodiversity indicator, lifetime tracking of individuals that does not require recapture and excessive handling will be valuable¹.

Priority actions

With these constraints in mind, we suggest priority actions for indicator development. First, the GBF's emphasis on connectivity (for example, in Goal A and Targets 2 and 3) should be met by grounding existing connectivity metrics in empirical movement data – beginning by giving stronger weight to protecting corridors that are directly estimated by observed animal movements. Second, new 'leading' movement-based metrics should be developed that flag rapid, actionable changes such as sudden increases in mortality or abrupt shifts in space use. Even metrics that do not become official indicators are valuable as a test bed for future development and will inform the data-sharing infrastructure required to support science–policy integration for movement data. Third, investment should be made in modelling and attribution studies that use movement data to identify drivers of declines and recoveries, and develop the statistical frameworks needed to detect aggregate change relative to a baseline. Most existing movement statistics and monitoring efforts are designed for individual-level and population-level inference, not indicator-scale reporting.

Coordination across regions, taxa, methods and repositories will be a consistent, major challenge⁴⁷, and Move BON is central to meeting this challenge by creating networks that connect researchers and practitioners across realms, ensuring interoperability and open-sharing commitments, and facilitating the collaborative scientific work that indicator development requires. Equitable sharing of knowledge, infrastructure, and funding – explicit GBF requirements and prerequisites for globally representative indicators – are core to Move BON's mandate.

Other advice can be drawn from general guidance on indicator development in ecology and conservation^{19,55,127,128} and the practical exercise of selecting indicators¹²⁹. Key recommendations include developing early trust by stakeholders through early and frequent involvement, establishing scientific and user committees, and assessing relevance, practicality and usefulness through the SMART-C (specific, measurable, achievable, relevant, timely and communicable) criteria framework¹²⁹ (Supplementary Information; Applying SMART-C to triage potential movement-based indicators).

Biodiversity indicators are essential tools for understanding and acting on changes to nature. Movement data offer spatiotemporally explicit insights into how species interact with their environments as well as intraspecific and interspecific interactions, how connectivity is maintained, and where population stress is emerging – signals that other data cannot easily capture. Integrating movement-based indicators into GBF reporting would strengthen the accuracy, relevance and timeliness of indicators, enabling more proactive and effective conservation worldwide.

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